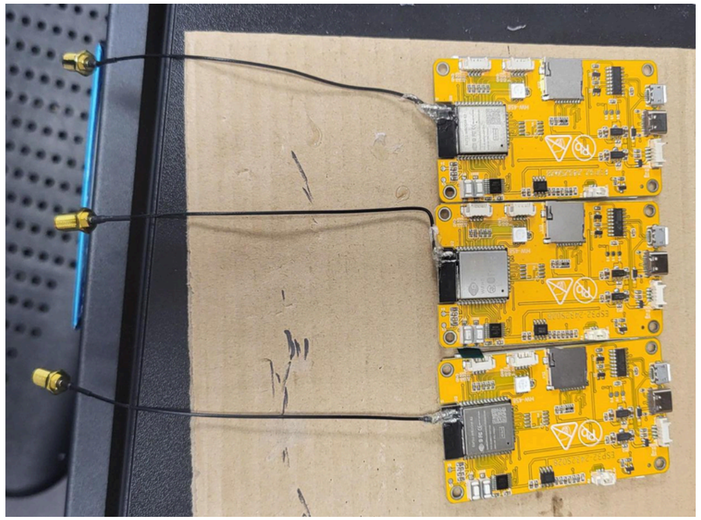
Air Sensor Progress Report

Loisaida - Sensor Case Team
Allegra Wong, Imtiaz Ahmed, Obinna Duru,
Terry Lin, Ayden Chiu

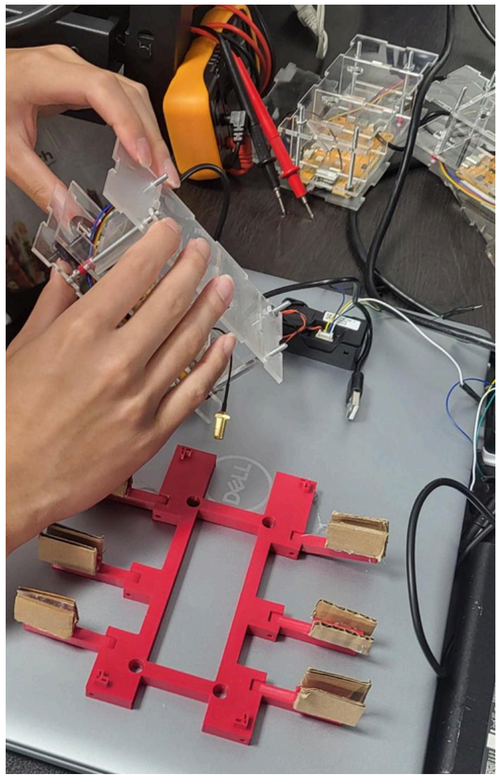
Week 2

Laser Cut Version Assembly

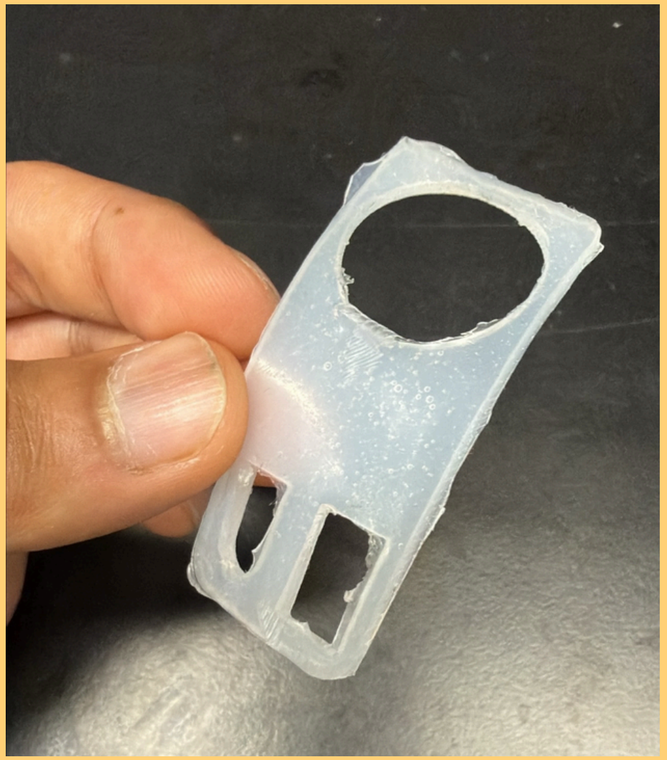


While we are designing a 3D printable case, we are also finishing assembly of 10 laser cut cases.

- resoldered antennas to ESPs
- used the assembly jig to assemble the laser cut versions
- ESPs were also flashed, tested, and troubleshot



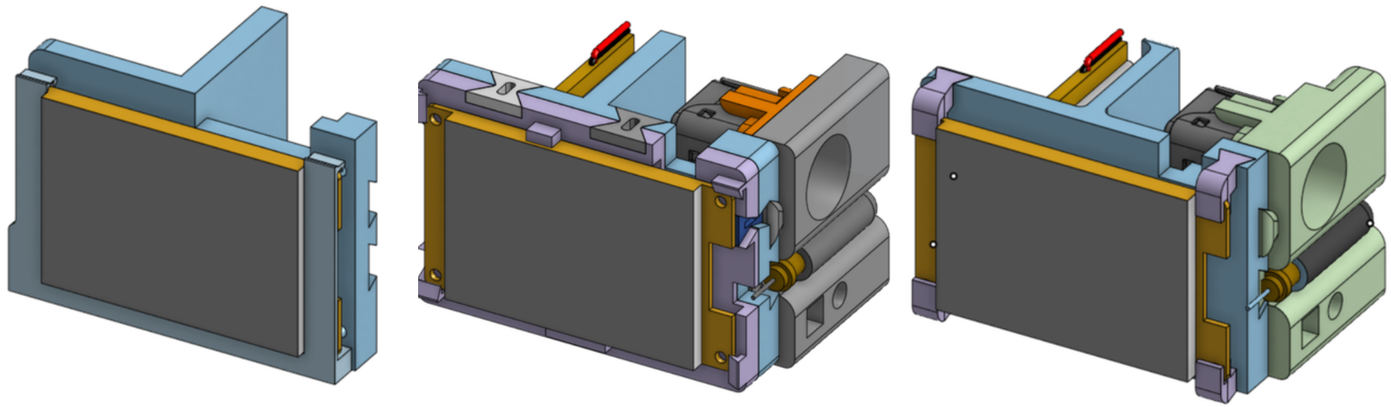
Testing cast silicone



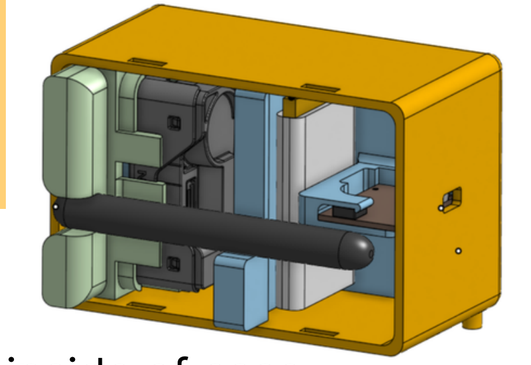
3D Modeling

Case without screws

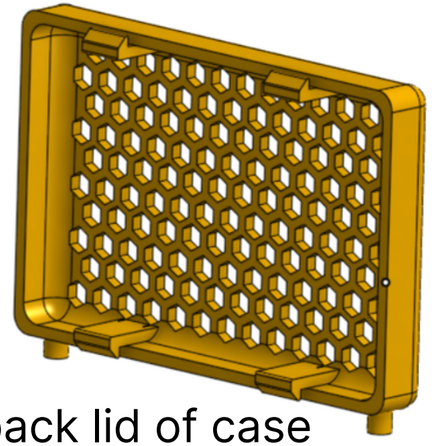
- Cantilever to attach case together
- snap fits and dovetails to hold parts together
- Air vents in the back plate
  - 3 different versions



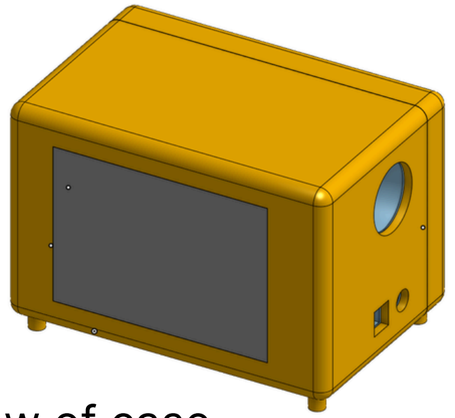
progression of screen mounts



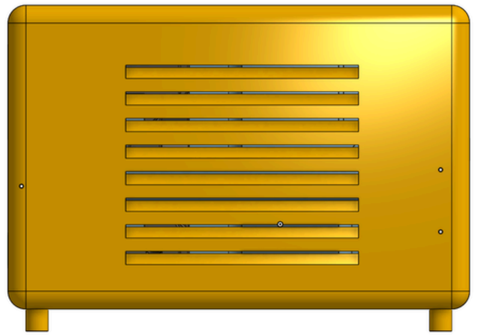
inside of case

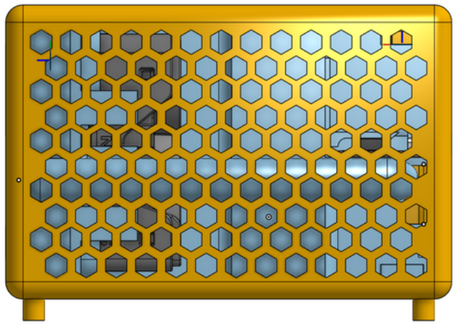


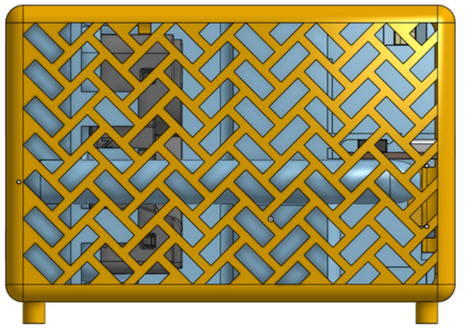
back lid of case



front iso view of case



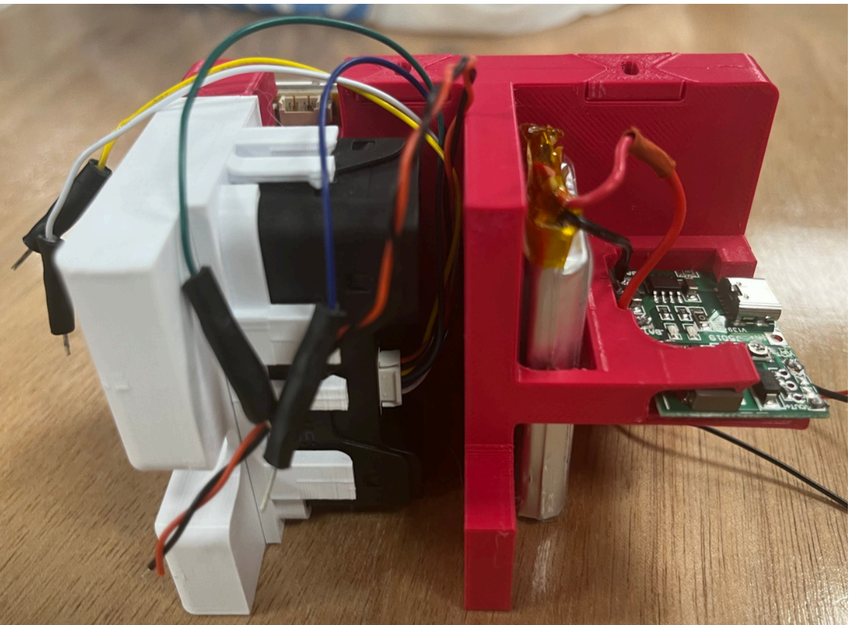


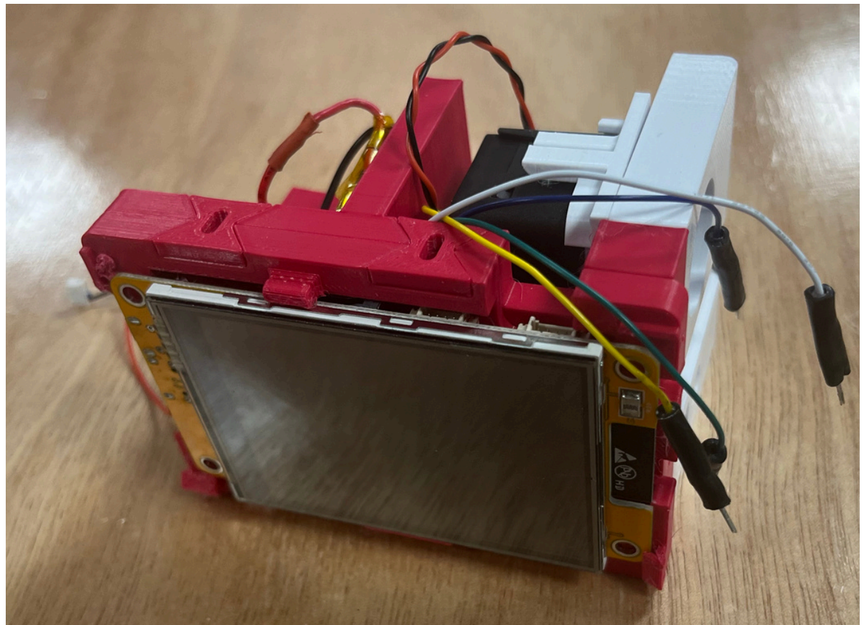


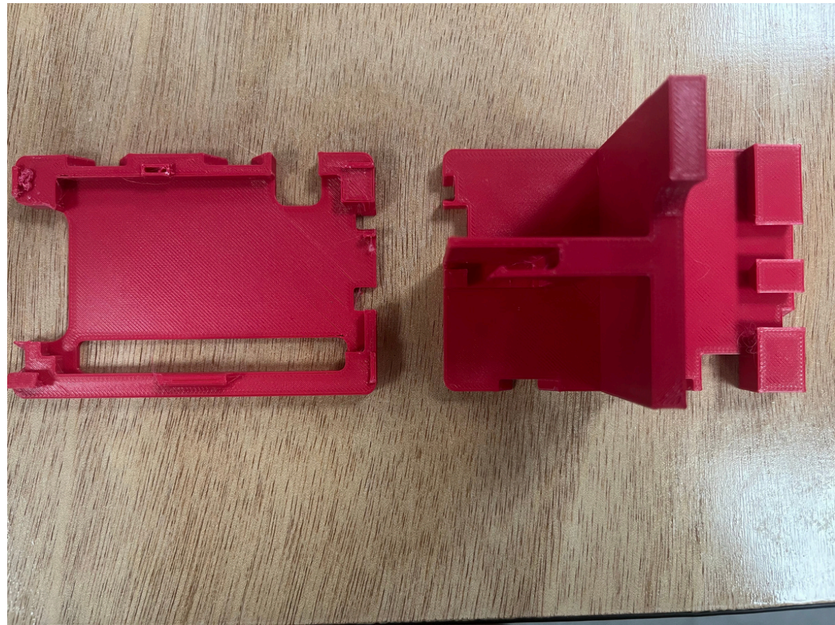
3D Prints

First skeleton and screen mount print

- screen mount uses cantilever snap fit to hold screen → broke
- Power board and battery don’t fit well into the skeleton
- dovetails attach screen mount to skeleton → replace snap fit of screen mount with dovetails



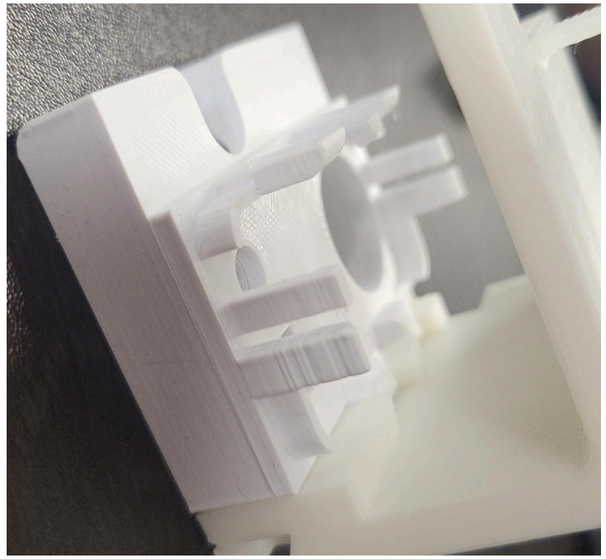




3D Prints

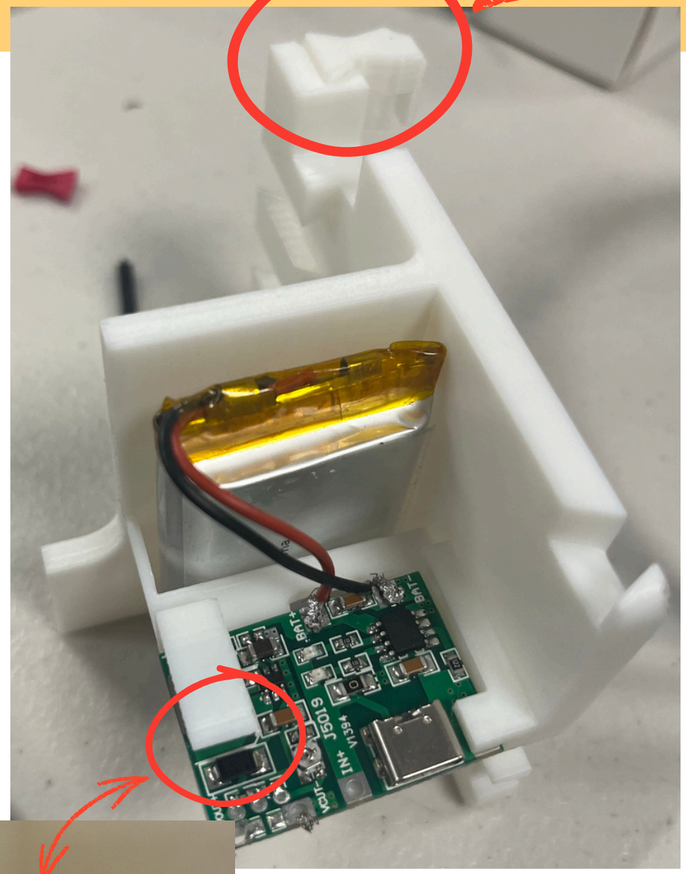
Second skeleton-dovetail print

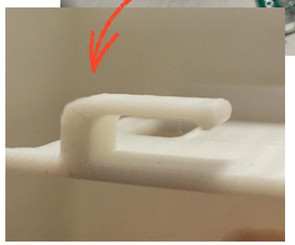
- Powerboard and battery fit perfectly now
- Dovetails are able to hold the screen without breaking
- Tolerances need to be adjusted



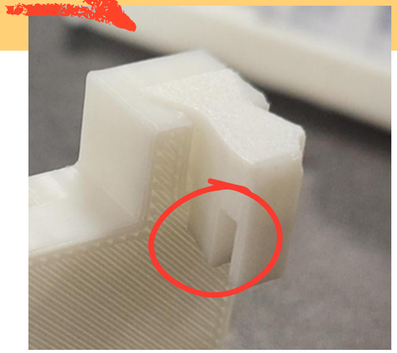
make antenna mount and sensor mount → easier assembly

adjust dovetail tolerance and add “stopper” in skeleton

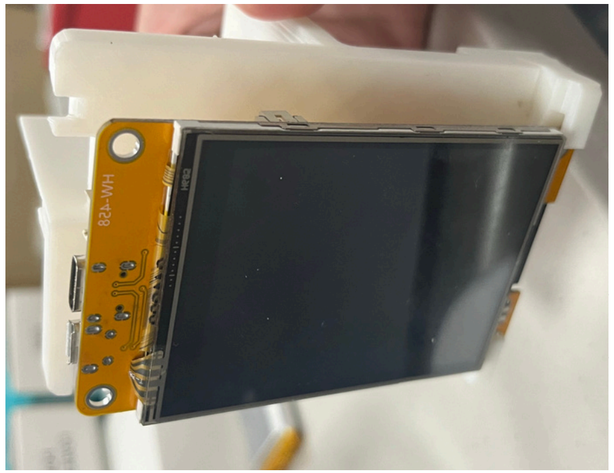




snap fit arm to hold power board in place



sloted piece holds esp, dovetail connects to skeleton



Next Steps

Manufacturing

- print and assemble next iteration of 3D-printed case and skeleton

Testing

- test functionality of electronics during daily use
- identify design flaws to optimize assembly speed, sensor airflow management, size

Procurement

- order laser cuttable rubber material to create gaskets to restrict air flow

THANK YOU